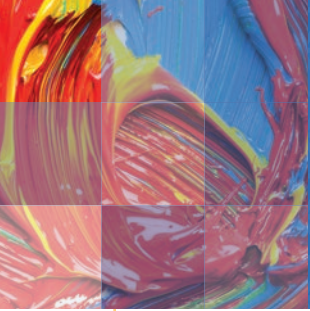




Domain in Linear Programming

Linear programming problems typically have whole number solutions.

If the vertex point that optimizes the objective function has coordinates that are not both integers, check integer values that are closest to the vertex point and that are still in the feasible region.

EXAMPLE

Brew Ha Ha Coffee & Tea Co. makes two types of handmade coffee mugs: matte and glazed. It takes 14 min on the pottery wheel and 20 min of kiln time to make each matte mug, and 15 min of wheel time and 30 min of kiln time to make each glazed mug. The company has reserved 300 min of time on the pottery wheel and 500 min of kiln time from a local potter. Brew Ha Ha charges \$12 for matte mugs and \$14 for glazed mugs.

A How many of each type of mug should Brew Ha Ha make to produce the maximum revenue?

SOLUTION

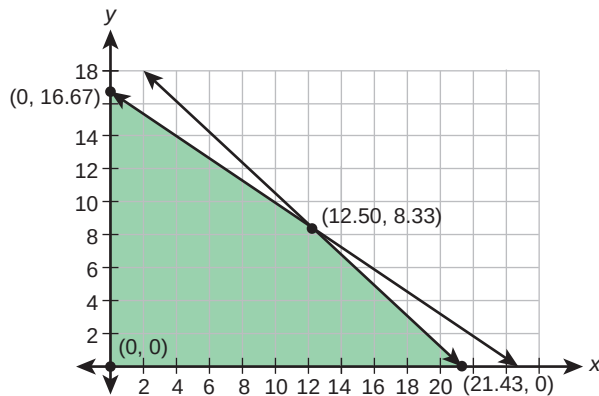
Step 1 Let x = number of matte mugs and y = number of glazed mugs.

Step 2 Since the company charges \$12 for each matte mug and \$14 for each glazed mug, the objective function is $R = 12x + 14y$. The company wants to maximize revenue, R .

Step 3 Write inequalities for the constraints.

$$\begin{cases} 14x + 15y \leq 300 & \text{no more than 300 min wheel time} \\ 20x + 30y \leq 500 & \text{no more than 500 min kiln time} \\ x \geq 0 & \text{number of matte mugs cannot be negative} \\ y \geq 0 & \text{number of glazed mugs cannot be negative} \end{cases}$$

Step 4



Step 5 The vertex points of the feasible region are $(0, 0)$, $(0, 16.67)$, $(12.50, 8.33)$, and $(21.43, 0)$.

Step 6 Evaluate the objective function at each vertex point.

Vertex point	$12x + 14y$	R
$(0, 0)$	$12 \cdot 0 + 14 \cdot 0$	\$0
$(0, 16.67)$	$12 \cdot 0 + 14 \cdot 16.67$	\$233.38
$(12.50, 8.33)$	$12 \cdot 12.50 + 14 \cdot 8.33$	\$266.62
$(21.43, 0)$	$12 \cdot 21.43 + 14 \cdot 0$	\$257.16

Since the maximum revenue occurs at a point $(12.50, 8.33)$ that contains nonintegers, try whole-number values near that point that are still within the constraints. Substituting $(12, 8)$ into the constraints shows that the company would use 288 h on the wheel and 480 h of kiln time, which is within the constraints. This vertex point yields revenue of \$256. However, this value is below \$257.16, which is the revenue created by vertex point $(21.43, 0)$. So round the x -value down to a whole number and determine the revenue there. Substituting $(21, 0)$ into the objective function yields \$252, which is below the \$256 revenue provided by $(12, 8)$. Thus, the company should make 12 matte mugs and 8 glazed mugs to maximize revenue.

B What is the maximum revenue?

SOLUTION

The maximum revenue is \$256.

C If the company decides to sell both types of mugs for \$13, how many of each should the company now make to maximize revenue?

SOLUTION

The feasible region and the vertex points are the same, but the objective function becomes $R = 13x + 13y$.

Vertex point	$13x + 13y$	R
$(0, 0)$	$13 \cdot 0 + 13 \cdot 0$	\$0
$(0, 16.67)$	$13 \cdot 0 + 13 \cdot 16.67$	\$216.71
$(12.50, 8.33)$	$13 \cdot 12.50 + 13 \cdot 8.33$	\$270.79
$(21.43, 0)$	$13 \cdot 21.43 + 13 \cdot 0$	\$278.59

The maximum revenue now occurs at $(21.43, 0)$, so check $(21, 0)$. This yields revenue of \$273, which is still the maximum revenue, so at the new prices the company should make 21 matte mugs and no glazed mugs.

D What is the adjusted maximum revenue?

SOLUTION

The adjusted maximum revenue is \$273. ■